STUDYNEST | CLASS 11 PHYSICS

MARKING SCHEME AND MODEL ANSWERS

Chapter 4: Laws of Motion

Maximum Marks: 40 | Suggested time: 60 minutes

Use after attempting the paper. Accept equivalent correct methods.

Suggested mark allocation:

1 mark: Correct fact, formula, unit or direction.
2 marks: Principle/formula (1); explanation, working or result (1).
3 marks: Relation (1); working/reasoning (1); final result and unit (1).
4 marks: Principle (1); setup/equation (1); working (1); result/interpretation (1).
Mark splits are a guide; award partial credit for valid steps.

Q1 - 1 mark

Mass.

Q2 - 1 mark

F=ma.

Q3 - 1 mark

No.

Q4 - 1 mark

p=mv.

Q5 - 1 mark

Zero.

Q6 - 1 mark

Opposes relative sliding.

Q7 - 1 mark

Inward resultant of actual forces.

Q8 - 1 mark

mu_sN.

Q9 - 2 marks

4 m/s^2.

Q10 - 2 marks

-3 kg m/s.

Q11 - 2 marks

98 N.

Q12 - 2 marks

24 N.

Q13 - 2 marks

10 N s.

Q14 - 2 marks

Water exerts a forward reaction.

Q15 - 3 marks

4 m/s^2.

Q16 - 3 marks

-60 N.

Q17 - 3 marks

5 m/s^2; 86.6 N.

Q18 - 3 marks

20 N inward.

Q19 - 4 marks

a=5 m/s^2; v=15 m/s.

Q20 - 4 marks

Deltap=-10 N s; force=-100 N.